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Lab 5 — Spatial Filtering: Smoothing and Sharpening

Objective: Apply spatial filters to smooth and sharpen an image, then compare their effects.

Tools: GoogleCollab, Python, OpenCV, NumPy, and Matplotlib

Task 1 — 7×7 box filter

Replaced each pixel with the average of its 7×7 neighbourhood. This smoothed the image and softened edges.

```
kernel = np.ones((7, 7), dtype=np.float32) / 49

smoothed = cv2.filter2D(image, -1, kernel)

fig, ax = plt.subplots(1, 2, figsize=(12, 5))

ax[0].imshow(image) ax[0].set_title("Original Image")

ax[1].imshow(smoothed) ax[1].set_title("7×7 Box Filter")

for a in ax: a.axis("off")

plt.tight_layout()

plt.show()
```

Task 2 — 5×5 and 21×21 Gaussian filters

Smoothed the image using weighted averages, giving nearby pixels more weight. The **21×21 filter produced stronger blur** than the 5×5 filter.

```
gaussian_5 = cv2.GaussianBlur(image, (5, 5), sigmaX=0)

gaussian_21 = cv2.GaussianBlur(image, (21, 21), sigmaX=0)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))


ax[0].imshow(image)
```

```
ax[0].set_title("Original Image")

ax[1].imshow(gaussian_5)

ax[1].set_title("5×5 Gaussian Filter")

ax[2].imshow(gaussian_21)

ax[2].set_title("21×21 Gaussian Filter")

for a in ax:

   a.axis("off")

plt.tight_layout()

plt.show()
```

Task 3 — 3×3 Laplacian sharpening

Detected rapid brightness changes, such as edges. Subtracting the Laplacian from the original image strengthened these edges.

```
image_float = image.astype(np.float32) / 255.0

laplacian = cv2.Laplacian( image_float, cv2.CV_32F, ksize=3 )

sharpened = np.clip(image_float - laplacian, 0, 1)

max_abs = np.max(np.abs(laplacian))

if max_abs > 0: laplacian_display = 0.5 + laplacian / (2 * max_abs) else:
laplacian_display = np.full_like(laplacian, 0.5)

fig, ax = plt.subplots(1, 3, figsize=(15, 5))

ax[0].imshow(image) ax[0].set_title("Original Image")

ax[1].imshow(laplacian_display) ax[1].set_title("3×3 Laplacian Response")

ax[2].imshow(sharpened) ax[2].set_title("Sharpened Image")

for a in ax: a.axis("off")

plt.tight_layout() plt.show()
```